```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.UI;
using Random = UnityEngine.Random;
using TMPro;

public enum GameState { dealerTurn, playerTurn, endGame }

public class GameController : MonoBehaviour
{
    public GameState gameState;

    [Header("UI")]
        public GameObject hitButton;
        public GameObject standButton;
        public GameObject restartButton;
        public GameObject quitButton;
        public TextMeshProUGUI endGameText;
    [Space(10)]

    [Header("Cards")]
        public GameObject dealerHandObject;
        public GameObject playerHandObject;
        public List<Card> deck = new List<Card>();
        public List<Card> dealerHand = new List<Card>();
        public List<Card> playerHand = new List<Card>();

    public int dealerHandValue;
    public int playerHandValue;

    float playerCardOffsetX = 0.0f;
    float playerCardOffsetY = 0.0f;
    float dealerCardOffsetX = 0.0f;
    float dealerCardOffsetY = 0.0f;

    public void Start()
    {
        foreach (Card card in deck)
            card.isSpawned = false;

        DrawPlayer(2);
        DrawDealer(2);
        DeckVisualUpdater(playerHand, playerHandObject);
        DeckVisualUpdater(dealerHand, dealerHandObject);
        endGameText.text = "";
        gameState = GameState.playerTurn;
    }

    public void Update()
    {
        CheckAce();

        if (playerHandValue >= 21 || dealerHandValue > 21)
        {
            DeckVisualUpdater(playerHand, playerHandObject);
            DeckVisualUpdater(dealerHand, dealerHandObject);
            gameState = GameState.endGame;
        }

        if (gameState == GameState.playerTurn)
        {
            hitButton.SetActive(true);
            standButton.SetActive(true);
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    }
    else if (gameState == GameState.dealerTurn)
    {
        hitButton.SetActive(false);
        standButton.SetActive(false);
        StartCoroutine(DealerTurn());
    }
    else
        EndGame();
}

public void DrawPlayer(int amount)
{
    for (int i = 0; i < amount; i++)
    {
        int randomCardNumber = Random.Range(0, deck.Count);
        Card randomCard = deck[randomCardNumber];
        deck.RemoveAt(randomCardNumber);
        playerHand.Add(randomCard);
        if (randomCard.value == 1)
            playerHandValue += randomCard.aceValue;
        else
            playerHandValue += randomCard.value;
    }
}

public void DrawDealer(int amount)
{
    for (int i = 0; i < amount; i++)
    {
        int randomCardNumber = Random.Range(0, deck.Count);
        Card randomCard = deck[randomCardNumber];
        deck.RemoveAt(randomCardNumber);
        dealerHand.Add(randomCard);
        if (randomCard.value == 1)
            dealerHandValue += randomCard.aceValue;
        else
            dealerHandValue += randomCard.value;
    }
}

public void PlayerTurnHit()
{
    DrawPlayer(1);
    DeckVisualUpdater(playerHand, playerHandObject);
    gameState = GameState.dealerTurn;
}

public void PlayerTurnStand()
{
    DeckVisualUpdater(playerHand, playerHandObject);

    if (dealerHandValue >= 16)
        gameState = GameState.endGame;
    else
        gameState = GameState.dealerTurn;
}

public IEnumerator DealerTurn()
{
    yield return new WaitForSeconds(1f);
    if (dealerHandValue < 16)
        DrawDealer(1);
```

```
        DeckVisualUpdater(dealerHand, dealerHandObject);
        gameState = GameState.playerTurn;
    }

    public void EndGame()
    {
        hitButton.SetActive(false);
        standButton.SetActive(false);
        restartButton.SetActive(true);
        quitButton.SetActive(true);
        DeckVisualUpdater(playerHand, playerHandObject);
        DeckVisualUpdater(dealerHand, dealerHandObject);

        if (playerHandValue > 21)
            endGameText.text = "You Lose";
        else if (dealerHandValue > 21)
            endGameText.text = "You Win";
        else
        {
            if (playerHandValue > dealerHandValue)
                endGameText.text = "You Win";
            else if (dealerHandValue > playerHandValue)
                endGameText.text = "You lose";
            else
                endGameText.text = "Tie";
        }
    }

    public void CheckAce()
    {
        foreach (Card card in playerHand)
        {
            if (card.value == 1 && playerHandValue > 21 && !card.aceValueCheck)
            {
                card.aceValueCheck = true;
                playerHandValue -= card.aceValue;
                playerHandValue += card.value;
            }
        }

        foreach (Card card in dealerHand)
        {
            if (card.value == 1 && dealerHandValue > 21 && !card.aceValueCheck)
            {
                card.aceValueCheck = true;
                dealerHandValue -= card.aceValue;
                dealerHandValue += card.value;
            }
        }
    }

    public void DeckVisualUpdater(List<Card> hand, GameObject handHolder)
    {
        foreach (Card card in hand)
        {
            if (!card.isSpawned)
            {
                GameObject spawnedCard = Instantiate(card.cardObject);
                spawnedCard.transform.SetParent(handHolder.transform);
                if (handHolder.tag == "Player")
                {
                    spawnedCard.transform.localPosition = new
Vector3(playerCardOffsetX, playerCardOffsetY, 0);
                    playerCardOffsetX += 0.5f;
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```
                    playerCardOffsetY += 0.1f;
                }
                else
                {
                    spawnedCard.transform.localPosition = new
Vector3(dealerCardOffsetX, dealerCardOffsetY, 0);
                    dealerCardOffsetX += 0.5f;
                    dealerCardOffsetY += 0.1f;
                }

                card.isSpawned = true;

                if (handHolder.tag != "Player")
                    if (spawnedCard.transform.position.x != 0)
                        spawnedCard.transform.rotation = new Quaternion(0, 0,
180, 0);
            }
        }

        if (gameState == GameState.endGame)
        {
            for (int i = 0; i < handHolder.transform.childCount; i++)
            {
                if (handHolder.transform.GetChild(i).transform.rotation.z != 0)
                    handHolder.transform.GetChild(i).transform.rotation = new
Quaternion(0, 0, 0, 0);
            }
        }
    }
}
```

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;

[CreateAssetMenu(fileName = "Card", menuName = "ScriptableObjects/Card")]
public class Card : ScriptableObject
{
    // Card asset by Game Asset Studio:
https://assetstore.unity.com/packages/3d/props/tools/free-playing-cards-pack-
154780
    [Header("Card Info")]
        public int value;
        public int aceValue;
        public bool aceValueCheck;
        public bool isSpawned;
        public GameObject cardObject;
}
```

```
using System.Collections;
using System.Collections.Generic;
using UnityEngine;
using UnityEngine.SceneManagement;

public class SceneController : MonoBehaviour
{
    public void ToGame()
    {
        SceneManager.LoadScene("Game");
    }

    public void ToTitle()
    {
        SceneManager.LoadScene("Title");
    }

    public void RestartScene()
    {
        SceneManager.LoadScene(SceneManager.GetActiveScene().buildIndex);
    }
}
```